

## MAZATLAN, Mexico

WMO: 764580

Lat: 23.2172N

Lon: 106.4111W

Elev: 10

StdP: 14.69

Time Zone: -7.00 (MEC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 57.0       | 58.9 | 46.5                          | 46.8 | 63.7 | 49.9 | 53.2 | 63.7 | 15.4                  | 71.7 | 13.8 | 73.8 | 3.3                   | 340  | 0.293 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 10.1                   | 91.7            | 81.1 | 90.9 | 80.7 | 90.2 | 80.4 | 83.0                | 89.3 | 82.3 | 88.8 | 81.7 | 88.3 | 6.9                  | 270  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 81.1                            | 161.8 | 87.8 | 80.5 | 158.1 | 87.5 | 79.7 | 154.4 | 87.1 | 47.0          | 89.7 | 46.1 | 89.0 | 45.4 | 88.4 | 88.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 14.1              | 12.3 | 11.1 | DB                         | 54.2 | 93.9               | 4.7 | 2.3                                                | 50.8 | 95.5       | 48.1 | 96.9       | 45.4 | 98.2       | 42.0 | 99.8 |
|                   |      |      | WB                         | 48.2 | 85.7               | 4.2 | 1.3                                                | 45.2 | 86.7       | 42.7 | 87.4       | 40.4 | 88.2       | 37.3 | 89.2 |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 77.5   | 68.8  | 69.6  | 70.4  | 73.0  | 78.6  | 83.7  | 84.6  | 84.6  | 84.1  | 82.8  | 77.8  | 71.2  |
|                                                                        | DBStd       | 6.81   | 3.37  | 3.53  | 3.01  | 2.57  | 2.87  | 2.21  | 2.05  | 2.10  | 2.10  | 2.67  | 2.87  | 3.62  |
|                                                                        | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 16     | 8     | 3     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 4     |
|                                                                        | CDD50       | 10025  | 582   | 549   | 632   | 689   | 885   | 1012  | 1073  | 1073  | 1023  | 1016  | 835   | 656   |
|                                                                        | CDD65       | 4567   | 125   | 132   | 168   | 239   | 420   | 562   | 608   | 608   | 573   | 551   | 385   | 196   |
|                                                                        | CDH74       | 47622  | 412   | 467   | 577   | 1168  | 3810  | 6946  | 8045  | 8079  | 7369  | 6621  | 3229  | 899   |
|                                                                        | CDH80       | 18221  | 15    | 25    | 28    | 82    | 921   | 2815  | 3723  | 3781  | 3255  | 2665  | 842   | 69    |
| (14) Wind                                                              | WSAvg       | 5.0    | 4.6   | 4.8   | 5.1   | 5.3   | 5.3   | 5.5   | 5.4   | 5.4   | 5.3   | 4.7   | 4.5   | 4.4   |
| (15) Precipitation                                                     | PrecAvg     | 27.8   | 0.7   | 0.4   | 0.2   | 0.0   | 0.0   | 1.3   | 6.2   | 9.3   | 9.6   | 2.7   | 0.8   | 0.6   |
|                                                                        | PrecMax     | 48.7   | 5.4   | 4.8   | 4.3   | 0.2   | 1.3   | 6.9   | 20.1  | 23.6  | 20.0  | 9.7   | 5.7   | 2.8   |
|                                                                        | PrecMin     | 14.8   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   | 0.8   | 0.4   | 0.0   | 0.0   | 0.0   |
|                                                                        | PrecStd     | 8.7    | 1.3   | 0.9   | 0.7   | 0.0   | 0.2   | 1.8   | 4.8   | 5.3   | 5.0   | 2.7   | 1.4   | 0.8   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 81.8  | 82.5  | 82.7  | 84.0  | 88.5  | 91.4  | 92.5  | 92.6  | 92.2  | 91.8  | 89.1  | 84.0  |
|                                                                        |             | MCWB   | 68.9  | 70.8  | 69.7  | 72.0  | 76.2  | 79.5  | 81.0  | 82.0  | 81.7  | 81.3  | 77.4  | 71.1  |
|                                                                        | 2%          | DB     | 79.5  | 80.2  | 80.3  | 82.1  | 86.8  | 90.2  | 91.4  | 91.6  | 91.0  | 90.4  | 87.2  | 81.8  |
|                                                                        |             | MCWB   | 68.9  | 70.6  | 69.4  | 71.5  | 75.7  | 79.0  | 80.5  | 81.5  | 81.4  | 80.7  | 76.7  | 71.1  |
|                                                                        | 5%          | DB     | 78.0  | 78.6  | 78.7  | 80.8  | 85.5  | 89.1  | 90.5  | 90.7  | 90.2  | 89.4  | 85.8  | 80.4  |
|                                                                        |             | MCWB   | 68.5  | 69.6  | 68.9  | 71.0  | 75.3  | 78.6  | 80.2  | 81.0  | 81.1  | 80.0  | 75.7  | 70.8  |
|                                                                        | 10%         | DB     | 76.5  | 77.0  | 77.3  | 79.5  | 84.4  | 88.2  | 89.5  | 89.8  | 89.3  | 88.4  | 84.4  | 78.9  |
|                                                                        |             | MCWB   | 67.8  | 68.5  | 68.1  | 70.3  | 74.8  | 78.2  | 79.8  | 80.6  | 80.7  | 79.3  | 74.9  | 69.9  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 72.1  | 73.6  | 72.8  | 74.6  | 78.6  | 81.8  | 83.5  | 84.1  | 83.7  | 83.4  | 79.9  | 74.6  |
|                                                                        |             | MCDB   | 78.3  | 79.7  | 79.7  | 81.6  | 86.0  | 88.3  | 90.3  | 90.3  | 89.6  | 89.4  | 87.0  | 80.7  |
|                                                                        | 2%          | WB     | 70.9  | 72.1  | 71.5  | 73.1  | 77.3  | 80.5  | 82.2  | 82.8  | 82.8  | 82.2  | 78.3  | 73.4  |
|                                                                        |             | MCDB   | 77.3  | 78.4  | 78.0  | 79.8  | 84.5  | 88.1  | 89.2  | 89.5  | 88.9  | 88.4  | 85.0  | 80.0  |
|                                                                        | 5%          | WB     | 69.7  | 70.8  | 70.3  | 72.2  | 76.4  | 79.7  | 81.3  | 82.0  | 82.1  | 81.4  | 77.3  | 72.2  |
|                                                                        |             | MCDB   | 76.2  | 77.0  | 76.6  | 79.0  | 83.5  | 87.4  | 88.5  | 88.9  | 88.4  | 87.7  | 83.8  | 78.3  |
|                                                                        | 10%         | WB     | 68.6  | 69.4  | 69.2  | 71.3  | 75.7  | 79.0  | 80.6  | 81.4  | 81.5  | 80.5  | 76.3  | 70.9  |
|                                                                        |             | MCDB   | 75.0  | 75.4  | 75.5  | 78.1  | 82.7  | 86.7  | 88.0  | 88.3  | 87.7  | 86.9  | 82.8  | 77.0  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 14.5  | 13.4  | 12.9  | 12.5  | 10.9  | 8.3   | 9.7   | 10.1  | 9.7   | 10.8  | 12.9  | 14.0  |
|                                                                        | 5% DB       | MCDBR  | 15.5  | 14.4  | 13.9  | 13.4  | 11.0  | 8.9   | 10.3  | 10.8  | 10.5  | 10.8  | 13.3  | 14.9  |
|                                                                        |             | MCWBR  | 9.0   | 8.3   | 7.4   | 7.0   | 5.3   | 4.4   | 5.2   | 5.3   | 5.1   | 5.2   | 7.4   | 8.5   |
|                                                                        | 5% WB       | MCDBR  | 14.4  | 13.0  | 12.6  | 12.8  | 10.3  | 8.3   | 9.8   | 10.0  | 9.7   | 10.1  | 12.3  | 13.5  |
|                                                                        |             | MCWBR  | 8.9   | 7.9   | 7.3   | 7.4   | 5.5   | 4.8   | 5.6   | 5.4   | 5.4   | 5.4   | 7.6   | 8.1   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.320 | 0.328 | 0.314 | 0.349 | 0.399 | 0.432 | 0.440 | 0.420 | 0.421 | 0.369 | 0.333 | 0.319 |
|                                                                        | taud        |        | 2.596 | 2.550 | 2.596 | 2.453 | 2.371 | 2.352 | 2.376 | 2.439 | 2.397 | 2.578 | 2.625 | 2.631 |
|                                                                        | Ebn at Noon |        | 302   | 305   | 313   | 301   | 284   | 272   | 270   | 277   | 276   | 288   | 295   | 298   |
|                                                                        | Edh at Noon |        | 28    | 31    | 31    | 36    | 40    | 40    | 39    | 36    | 37    | 30    | 27    | 26    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1413  | 1721  | 2107  | 2355  | 2453  | 2210  | 2000  | 1928  | 1714  | 1715  | 1519  | 1332  |
|                                                                        | RadStd      |        | 108   | 144   | 69    | 120   | 80    | 92    | 101   | 67    | 81    | 81    | 84    | 75    |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | +1.15       | N/A   | N/A   | N/A   |

Nomenclature: See separate page